

3-Level Geometric Modeler: CSG, Subdivision Trees and Boundary Representation

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I. INTRODUCTION

Solid modeling is distinguished from related areas of geometric modeling and computer graphics by its emphasis on physical fidelity. Together, the principles of geometric and solid modeling form the foundation of computer-aided design and in general support the creation, exchange, visualization, animation, interrogation, and annotation of digital models of physical objects.

We will describe a polyhedron as three sets : (V, E, F) . V represents the set of vertices of the polyhedron. E represents the set of edges of the polyhedron. F represents the set of faces of the polyhedron.

II. DEFINITIONS

2.1 Regularized set

A set $S \subset \mathbb{R}^3$ is a regularized set iff $S = \overline{\text{int}(S)}$. All of our operations are on Regularized sets.

2.2 Regularized set operation on regularized sets A and B

- $A \cup^* B = \overline{\text{int}(A \cup B)}$
- $A \cap^* B = \overline{\text{int}(A \cap B)}$
- $A -^* B = \overline{\text{int}(A - B)}$
- $A^c * B = \overline{\text{int}(A^c)}$

2.3 Solid

A solid S is defined as a regularized subset of $\mathbb{R}^3$ ($s \subset \mathbb{R}^3$)

We define the set containing all solids as $\mathcal{S}$.

III. CSG OBJECTS

Just as a polyhedron can be described as the intersection of half spaces, a CSG (Constructive Solid Geometry) model describes a polyhedron as regularized set operation on primitives which are the simplest solid objects used for the representation.

3.1 Primitives

Primitives in CSG models are all semi-algebraic sets as follows:

1. Half Space: $H = \{(x, y, z) | a * x + b * y + c * z - d \geq 0\}$
2. Sphere : $S = \{(x, y, z) | (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 - r^2 \leq 0\}$.

3.2 Polyhedron

A polyhedron P is a finite solid that is the intersection of several half planes. We define the set containing all polyhedra as $\mathcal{P}$.

IV. SUBDIVISION TREES

1. Box: B is a box iff $B \subset \mathbb{R}^3$, $B = \{(x, y, z) | x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$, which is a regularized subset of $\mathbb{R}^3$

We define the set containing all boxes as $\mathcal{B}$.

2. Elementary Box for a polyhedron P :

- (a) Empty Box: a box B is an Empty Box iff $B \cap P = \emptyset$.
- (b) Full Box: a box B is a Full Box of Polyhedron P iff $B \cap P = B$

- (c) Vertex Box: a box B is a Vertex Box iff there exists only one vertex $v \in V \cap B$, s.t. $\forall e \in E$, if $e \cap B \neq \emptyset$, v bounds e , and $\forall f \in F$, if $f \cap B \neq \emptyset$, v bounds F .
 - (d) Edge Box: a box B is an Edge Box iff $V \cap B = \emptyset$ and there exists only one edge $e \in E$, $e \cap B \neq \emptyset$, $\forall f \in F$, if $f \cap B \neq \emptyset$, e bounds f .
 - (e) Face Box: a box B is a Face Box iff $\forall e \in E$, $e \cap B = \emptyset$, and there exists one face $f \in F$, and $f \cap B \neq \emptyset$.
3. Feature of a box corresponding to a vertex
 $\forall v \in V$, $\phi_0^*(B, v)$ is the coordinates of v if $v \in B$.
 $\Phi_0^*(B, P) = \{\phi_0^*(B, v) | v \in V\}$
 4. Feature of a box corresponding to an edge
 $\forall e \in E$, $\phi_1^*(B, e)$ is the coordinates of the two endpoints of e and $B \cap e$ which can be a segment or a vertex. $\Phi_1^*(B, P) = \{\phi_1^*(B, e) | e \in E\}$
 5. Feature of a box corresponding to a face
 $\forall f \in F$, $\phi_2^*(B, f)$ is the coordinates of the all vertices of f and $B \cap f$ which can be a polygon, an edge or a vertex.
 $\Phi_2^*(B, P) = \{\phi_2^*(B, f) | f \in F\}$
 6. Feature of a box corresponding to polyhedron P , the corresponding sets of P are (V, E, F)
 $\Phi^*(B, P) = (\Phi_0^*(B, P), \Phi_1^*(B, P), \Phi_2^*(B, P))$
 7. Subdivision tree : A subdivision tree in $\mathbb{R}^3$ is an octree, in which each node points to a box and we divide each box pointed by a inner node into 8 children pointing to 8 subboxes.
 8. Adequate subdivision tree for box B and polyhedron P :
 $T = SD(B_0, P)$ in which SD is a function on a box B and a polyhedron P , the root of the tree points to box B_0 . All leaves of T point to an elementary box of the polyhedron P .
 9. Boolean operations on two adequate subdivision trees $T_1 = SD(B_1, P_1)$ and $T_2 = SD(B_2, P_2)$: $T = M(BO, T_1, T_2)$ in which BO is a Boolean operation and T_1 and T_2 are adequate subdivision trees. The result of the function M is an adequate subdivision tree T .
if $B_1 \neq B_2$, $T = Null$.
if $B_1 = B_2 = B_0$, T is an adequate subdivision tree whose root points to B_0 . BO are Boolean operations as union ($\cup$), intersection ($\cap$) and difference ($-$).
- We can also define:
- $T_1 \cup T_2 = M(\cup, T_1, T_2)$
 - $T_1 \cap T_2 = M(\cap, T_1, T_2)$
 - $T_1 - T_2 = M(-, T_1, T_2)$
10. Subdivision Rules of a box B_0 corresponding to a polyhedron P :
 - (a)
 - (b)
 11. Boolean operation on an elementary box for two polyhedra P_1 and P_2
Suppose box B_e is an elementary box for both P_1 and P_2 . The the result of the Boolean operation on an an elementary box for two polyhedra P_1 and P_2 is an adequate subdivision tree.
We define a function m as: $T = m(BO, B_e, P_1, P_2)$ in which BO is a Boolean operation as $\cup$, $\cap$ and $-$, P_1 and P_2 are two polyhedra. The result of the function is an adequate subdivision tree. What is more, we define a generalized function m , when BO is a complement, which is also a boolean operation, one of the two polyhedron should be null (because the complement c is a unary operator). Arbitrarily we define $P_2 = Null$, then $T = m(^c, B_s, P_1)$.
 P_1 and P_2 are commutative in this function.

V. BOOLEAN OPERATION ON AN ELEMENTARY BOX B_e FOR TWO POLYHEDRA P_1 AND P_2

5.1 Union

1. If B_e is an empty box of either P_1 or P_2 (let us say P_1), the result $T = m(\cup, B_e, P_1, P_2)$ is a subdivision tree whose root points to B_e and has no children. B_s is an elementary box for P_2 .
2. if B_e is a full box of either P_1 or P_2 (let us say P_1), $T = m(\cup, B_e, P_1, P_2)$ is a subdivision tree whose root points to B_e and has no children. B_s is an elementary box for P_1 .
3. A polygon Box & A polygon Box:
 - (a) A polygon Box: if one half plane contains another (a belongs to b)
 - (b) A corner Box: if two segments crosses and the cross point lies within the Box (the intersection of a and b is not empty)
- (c) Not a simple Box: if there is no intersection between the two half planes, then the Box will be divided into 3 parts and finally we have to subdivide the Box until we get simple Boxes. (a and b are disjoint)
4. A polygon Box & A corner Box: This is the union of several half planes. Let us say that the half space containing this polygon is p. And the corner c is the intersection of half spaces $s_1, s_2, \dots, s_n$
 - (a) if p passes through the end point of corner c: a corner box or a half plane
5. A corner Box & corner Box: Let us suppose that we have corner c_1 and c_2 . c_1 is the intersection of half space $s_1, s_2, \dots, s_n$, while c_2 is the intersection of half space $t_1, t_2, \dots, s_n$.